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github.com/vishnugithubforpython

SUMMARY

I specialize in machine learning, data analysis, and backend development. With one year of experience at AiSpry, I have optimized data pipelines, built predictive analytics models, and delivered scalable AI-driven backend solutions. I was deployed onsite at the National Centre for Statistics and Information (NCSI), Oman Tax Authority, collaborating with government stakeholders on socio-economic prediction systems powered by geospatial intelligence.

TECHNICAL SKILLS

Languages: Python, SQL

Frameworks Tools: Django, Flask, FastAPI, React, TensorFlow, Keras, OpenCV, Pandas, NumPy, Matplotlib, Seaborn, scikit-learn, PySpark

Cloud DevOps: AWS S3, AWS EC2, Git, WinSCP, Ubuntu

Databases: PostgreSQL, SQL Server Management Studio, Firebase

Other Skills: Machine Learning, Deep Learning, Data Structures Algorithms, HLD, LLD, QGIS (Geospatial Data)

PROFESSIONAL EXPERIENCE

Ai Spry, Chennai: Associate Data Scientist

Dec 2024-Nov 2025

Client: North Western Railways

- Built an end-to-end audio intelligence system leveraging Google Gemini 2.5 Pro to transcribe Hindi railway announcements, identify operational keywords, and generate timestamps + frequency metrics for real-time insight.
- Designed an automated audio preprocessing and transcription pipeline powered by FFmpeg (PCM to mono conversion), SpeexDSP (audio enhancement), and Pydub (chunking) orchestrated via cron jobs for continuous processing.
- Integrated AWS S3 for scalable audio storage and PostgreSQL (AWS RDS) for structured metadata and analytics, enabling fast search, tracking, and audit of announcements.
- Skills: Python, Flask, FastAPI, React, JavaScript, AWS S3, AWS EC2, PostgreSQL, Google Gemini, Speech-to-Text, Keyword Spotting, Audio Processing (SpeexDSP)

Client: NCSI Oman (National Centre for Statistics and Information)

- Performed on-site Exploratory Data Analysis (EDA) on Oman Tax Authority datasets, generating automated profiling reports using Sweetviz, computing correlations and mutual information to uncover key drivers of demographic consumption patterns.
- Processed VIIRS night-time satellite imagery in QGIS, using Oman shapefiles, raster clipping, and masked extraction (H23V06 tiles) to isolate the country's night-light intensity zones.
- Computed zonal statistics to quantify illumination levels across administrative regions, enabling predictive modelling of household income and expenditure using night-light intensity as a socio-economic proxy.
- Skills: Python, QGIS, VIIRS Night-Time Data, Geospatial Analytics, EDA, Machine Learning

Saturn Pyro, Chennai: Backend Developer

July 2024 - Aug 2024

- Developed modules for MyESG Online, a web-based platform enabling organizations to record and manage Environmental, Social, and Governance (ESG) metrics and compliance data.
- Implemented user authentication, role-based access, and ESG data submission workflows using Django, ensuring secure onboarding and structured data capture.
- Integrated AWS S3 for reliable storage of ESG documents and evidence files, enabling scalability.

Excel R, Data Science Intern

Feb 2024 – Jun 2024

- Apple Stock Market Forecasting: A 30-Day Predictive Analysis

EDUCATION

B.Tech., Artificial Intelligence and Data Science
Anna University, KCG College of Technology

Graduating May 2024
8.34 CGPA